

Your First Python Project — venv & pip

Beginner quick-start • Create a project, make a virtual environment, install a package, run it • (Linux / Ubuntu)

1 • Make the project folder

```
# go somewhere you can find it, then make + enter a folder
cd ~/Desktop
mkdir my-first-python-app
cd my-first-python-app
python3 --version          # check Python works, e.g. Python 3.8.10
```

2 • Create the virtual environment (once)

```
python3 -m venv .venv
ls -a                # .venv is hidden — use -a to see it
```

⚠ **Ubuntu error?** If you see "ensurepip is not available ... install the python3-venv package", Ubuntu left out the venv module. Fix it once:

```
rm -rf .venv          # remove the half-made folder
sudo apt install python3-venv  # enter your laptop password
python3 -m venv .venv    # try again — now it works
```

3 • Activate it (every session)

```
source .venv/bin/activate
```

Your prompt now starts with `(.venv)` — that means you're inside the environment:

```
(.venv) you@laptop:~/Desktop/my-first-python-app$
```

i Inside the venv plain `python` and `pip` now work (no need for `python3` / `pip3`).

4 • Install a package with pip

```
pip install requests    # requests = Python's fetch/axios
pip list                # confirm it's installed
```

You'll end with: **Successfully installed requests-2.x.x ...**

5 · Write a tiny script

```
nano test.py          # opens a terminal editor
```

Type this, then save with **Ctrl+O** → **Enter**, exit with **Ctrl+X**:

```
import requests

response = requests.get("https://api.github.com")
print("Status code:", response.status_code)
print("It works! 🎉")
```

6 · Run it

```
python test.py
```

Status code: 200 → success! You created a venv, installed a package, and used it. 🎉

7 · Leave the environment

```
deactivate          # prompt loses the (.venv) prefix
```

Cheat sheet

Action	Command	Node.js equivalent
Create env (once)	<code>python3 -m venv .venv</code>	(npm auto-makes node_modules)
Enter env (each session)	<code>source .venv/bin/activate</code>	<code>nvm use</code>
Install a package	<code>pip install requests</code>	<code>npm install requests</code>
See installed packages	<code>pip list</code>	<code>npm list</code>
Save dependencies	<code>pip freeze > requirements.txt</code>	writes to package.json
Leave env	<code>deactivate</code>	(none)

🔔 **Remember:** One `.venv` per project — created once, activated every session, never committed to git.